Von Neumann entropy reveals alternative stable state architecture in intertidal community succession

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Abstract

Alternative stable states are central to community ecology, yet detecting them from field data remains difficult because conventional diversity metrics treat species as interchangeable. Here I applied Von Neumann entropy—computed on density matrices derived from species rank-correlation matrices—to quantify the effective dimensionality of community interaction structure in experimentally cleared intertidal plots in the Gulf of Maine. Using seven years of succession data (1996–2002) across four bays, five clearing sizes, and eight species, I showed that Von Neumann entropy declined consistently over succession (slope = −0.039 yr^{-1}, bootstrap 95% CI [−0.068, −0.003], $P = 0.013$), indicating self-organization of the species interaction network toward lower-dimensional structure. Von Neumann entropy also varied non-monotonically with clearing size, reaching a minimum at 2 m^2 clearings (normalized $S = 0.856$ vs 0.926 for controls; $P = 0.025$)—the scale where prior theory predicted divergent trajectories. Species-label permutation confirmed that the observed entropy reflected genuine interaction structure ($z = -6.4$ to −22.6, all $P < 0.001$ except the earliest post-clearing timepoint). Data-driven community detection revealed that bivalves acted as "swing species" whose network allegiance shifted across successional and spatial contexts. These results demonstrate that Von Neumann entropy of species interaction density matrices provides a principled, information-theoretic signature of community phase structure not reliably detected by conventional diversity indices.

Keywords density matrix; information theory; community assembly; species interaction networks; von Neumann Entropy

Introduction

The concept of alternative stable states—that ecological communities can occupy qualitatively distinct configurations under identical environmental conditions—has been central to community ecology since Lewontin (1969) and Sutherland (1974). Theoretical formalization of thresholds between alternative states followed rapidly (May 1977; Holling 1973), and subsequent reviews established alternative stable states as a general phenomenon across ecosystems (Scheffer et al. 2001; Beisner et al. 2003), though direct experimental evidence remained limited (Schröder et al. 2005). Rocky intertidal communities have served as model systems for understanding how predation (Paine 1966), physical stress gradients (Connell 1972; Menge and Sutherland 1987), and successional mechanisms (Connell and Slatyer 1977) interact to structure communities. In rocky intertidal systems the question takes a concrete form: do mussel beds and macroalgal canopies represent genuinely alternative attractors, or merely different points on a single successional trajectory? Petraitis and colleagues addressed this experimentally in the Gulf of Maine by clearing intertidal substrate at multiple spatial scales and tracking community reassembly (Petraitis and Latham 1999; Petraitis and Dudgeon 2004, 2005). Their design exploited a key theoretical prediction: clearing size should determine which attractor a community converges toward, with large clearings capable of pushing the system past a basin boundary that small clearings cannot reach. Subsequent work provided experimental confirmation of multiple community states (Petraitis et al. 2009) and a catastrophe-theoretic interpretation of the stability landscape (Petraitis and Dudgeon 2015a).

Detecting alternative stable states from empirical data, however, remains methodologically difficult (Petraitis and Dudgeon 2004; Petraitis and Hoffman 2010). Classical diversity metrics, such as Shannon entropy, Simpson's index and species richness, aggregate

abundances into scalar summaries that treat species as interchangeable tokens (reviewed in Magurran 2004). These metrics measure *distributional* entropy (how evenly abundance is spread) but not *structural* entropy (how organized the network of species co-occurrences is). Yet it is the structural dimension—which species covary with which, and how tightly—that distinguishes alternative stable states from variation within a single basin of attraction.

Here I introduce an information-theoretic approach to this problem. I computed the Von Neumann entropy of species correlation density matrices as a measure of the effective dimensionality of community interaction structure. Von Neumann entropy, originally developed in quantum statistical mechanics (von Neumann 1927), generalizes Shannon entropy to density matrices (Passerini and Severini 2009): given a correlation matrix R among n species, one forms the density matrix $\rho = R/\mathrm{tr}(R)$ and computes $S(\rho) = -\mathrm{tr}(\rho \log \rho)$. The result ranges from 0 (all variation along a single axis) to $\log_2(n)$ (uniform eigenvalues; species uncorrelated). While correlations do not imply direct interactions (Blanchet et al. 2020), patterns of species co-variation integrate both direct and indirect ecological effects and have long been used to infer emergent community structure (Legendre and Legendre 2012).

Network approaches have become central to understanding how species interaction architectures shape community dynamics (Bascompte 2010; Proulx et al. 2005). Von Neumann entropy has been applied to ecological contact networks (Brower 2020) and to define information-theoretic distance measures between complex networks (De Domenico and Biamonte 2016); its value on graph structures captures non-trivial topological features beyond degree statistics alone (Minello et al. 2019). I extend this logic, tracking Von Neumann entropy as a community-level ecological observable.

Using the 1996–2002 dataset from Petraitis and colleagues' clearing experiment on Swan's Island, Maine (Petraitis and Vidargas 2006), I addressed three questions: (1) Does Von Neumann entropy decline over succession, indicating self-organization? (2) Does it vary with clearing size, identifying the scale at which alternative trajectories diverge? (3) Does data-driven community detection recover groupings that differ from a priori taxonomic guilds?

Materials and methods

Study system and data

I analyzed the publicly available dataset from the long-term intertidal clearing experiment on Swan's Island, Gulf of Maine, USA (44°8-10′N, 68°25-29′W; Petraitis and Vidargas 2006). The experiment spanned four bays (Birch Cove, Moose Cove, Sandy Cove, Timber Cove), each containing three sites, sampled from 1996 to 2002. At each site, quadrats of five sizes (0 [uncleared control], 1, 2, 4, and 8 m^2) were established and monitored for community reassembly (see Petraitis and Dudgeon 2005 for experimental design).

Eight species were quantified: the gastropods *Littorina littorea* (LL), *Littorina obtusata* (LO), *Nucella lapillus* (NL), and *Tectura testudinalis* (TT); the macroalgae *Fucus vesiculosus* (FV) and *Ascophyllum nodosum* (AN); and the bivalves *Mytilus edulis* (ME) and *Modiolus modiolus* (MM). Missing values (limited to FV in 1996 and AN in 1997) were excluded; the six core species had complete records across all 588 observations. Because F. vesiculosus was not recorded in 1996 and A. nodosum was not recorded in 1997, complete 8-species records were unavailable before 1998. I therefore used a 6-species subset for the full 7-year temporal analysis

and the complete 8-species set for 1998–2002 and for the spatial gradient analysis. The dataset comprised 588 observations with $n = 84$ per size class when pooled across years.

Von Neumann entropy

For each stratum (year, clearing size, year × size, or bay), I computed the Spearman rank-correlation matrix R among all species. I then formed the density matrix $\rho = R/\mathrm{tr}(R)$ and computed:

$$S(\rho) = -\sum_i \lambda_i \log_2 \lambda_i$$

where λ_i were the eigenvalues of ρ. I reported both raw S (bits) and normalized $S/S_{max} = S/\log_2(n)$ for comparability across the 6- and 8-species subsets. Because Spearman correlation matrices estimated from finite samples are not guaranteed to be positive semidefinite, any negative eigenvalues of ρ were set to a negligible positive value (10^{-15}) before entropy computation. In practice, no negative eigenvalues of ρ occurred in any stratum; the clipping threshold was included for generalizability to datasets with higher species-to-sample ratios.

Guild structure evaluation and community detection

To evaluate whether a priori taxonomic guilds (Grazers: LL, LO, NL, TT; Macroalgae: FV, AN; Bivalves: ME, MM) captured the observed correlation structure, I applied Ward's hierarchical clustering to the correlation distance matrix ($1 - |r\sim ij\sim|$) to identify data-driven species groupings (k = 3 for 8 species, k = 2 for 6 species). Agreement between data-driven and a priori groupings was quantified as the fraction of species pairs where both methods agreed on same-group vs different-group membership.

Statistical validation

I assessed significance with three complementary approaches, each with 1,000 replicates. (1) *Bootstrap confidence intervals*: sites were resampled with replacement within each stratum. For the temporal trend, I bootstrapped the slope of S against year; P was computed as 1 minus the fraction of bootstrap slopes < 0. For the Size 0 vs Size 2 contrast, I bootstrapped the difference in S. (2) *Species-label permutation*: species labels were shuffled within each row, destroying correlation structure while preserving marginal abundance distributions. I compared the observed S to the null distribution to obtain z-scores. (3) *Row-label permutation*: year or size labels were shuffled across rows, testing whether the trajectory range and slope differed from null expectations. All analyses were performed in Python 3 using NumPy, SciPy, and pandas.

Results

Temporal self-organization

Normalized Von Neumann entropy of the 6-species correlation density matrix declined from S/Smax = 0.939 in 1996 to 0.853 in 2002, a 9.1% reduction in the effective dimensionality of community interaction structure over seven years, with a single non-monotonicity between 1996 and 1997 (Fig. 1a; Table 1). The bootstrapped temporal slope was -0.039 yr^{-1} (95% CI [−0.068, −0.003]; 98.7% of replicates negative; $P = 0.013$). Row-label permutation confirmed the trajectory was non-random: the observed temporal range (0.264) exceeded 99.2% of null permutations ($P = 0.008$) and the slope was more negative than all 1,000 null values ($P < 0.001$).

Species-label permutation confirmed that observed Von Neumann entropy reflected genuine interaction structure. In all years except 1996, observed S was significantly lower than

the null expectation ($z = -6.4$ to -22.5, all $P < 0.001$; Table 2). The single non-significant year was 1996 ($z = -1.5$, $P = 0.077$), which had the smallest sample ($n = 48$) and represented the earliest post-clearing timepoint, when community structure had not yet differentiated from random.

The 8-species analysis (1998–2002) showed a parallel but noisier decline, from S/S_{max} = 0.883 to 0.857, punctuated by a local increase in 2001 (0.906; Table 1).

Non-monotonic complexity along the clearing-size gradient

Von Neumann entropy varied non-monotonically across clearing sizes (Fig. 1b; Table 3). Uncleared controls had the highest normalized entropy (S/S_{max} = 0.926). Entropy decreased to a minimum at Size 2 (0.856), then increased at Size 4 (0.902) before declining at Size 8 (0.883). The Size 0 vs Size 2 difference was significant (bootstrap 95% CI [0.0004, 0.358]; $P = 0.025$). Row-label permutation confirmed spatial non-randomness ($P = 0.021$).

All five size classes showed genuine interaction structure in species-label permutation ($z = -10.2$ to -22.6, all $P < 0.001$; Table 2). Size 2 had the most negative z-score (-22.6), indicating the greatest departure from the unstructured null. The Year × Size interaction (Fig. S1) showed that the Size 2 minimum was consistent across years, with the deepest trough at Size 2 × 2001 (S/S_{max} = 0.683).

Emergent community structure

Hierarchical clustering identified three modules in the 8-species analysis that did not correspond to a priori guilds. Pairwise agreement ranged from 0.54 to 0.75 across strata (Fig. 2). Three consistent patterns emerged: (1) the small gastropods (*L. obtusata*, *N. lapillus*, *T.*

testudinalis) formed a stable module; (2) the macroalgae *F. vesiculosus* and *A. nodosum* consistently paired; and (3) the bivalves *M. edulis* and *M. modiolus* did not reliably cluster together, instead acting as "swing species" whose network allegiance shifted across strata (Tables 1, 3). *Littorina littorea* was similarly structurally distinct from the other gastropods.

Discussion

Self-organization as entropy decline

The consistent decline in Von Neumann entropy over succession (Fig. 1a) has a known interpretation: the effective dimensionality of the species interaction network decreased as the community matured. In 1996, shortly after clearing, species co-occurrence patterns were close to maximum entropy. By 2002, strong correlations had reduced the system to a lower-dimensional manifold. This is the information-theoretic signature of ecological self-organization: the entropy decline is not a consequence of analytical dimensionality reduction applied to the data, but an observable property of the community's own correlation structure. To my knowledge, ecological succession has not previously been formalized as dimensional reduction in species interaction space. This parallels Odum's (1969) prediction that ecosystem development involves increasing organization, but provides a formal information-theoretic measure rather than a qualitative expectation.

That 1996 was the only year where species-label permutation did not reach significance ($P = 0.077$) reinforces this interpretation: the earliest post-clearing community had not yet developed the interaction structure distinguishing it from random assemblage. Subsequent years

showed consistently large departures from the null ($z = -6.4$ to -22.5), documenting the emergence of structure in real time.

This finding complements but differs from what Shannon entropy would reveal. Shannon entropy measures whether species abundances are even; Von Neumann entropy measures whether species *relationships* are organized. Here, Shannon entropy computed on the same strata showed a marginally non-significant decline (slope = −0.040/yr, P = 0.059 by permutation), with substantial interannual fluctuation (e.g. a spike in 1998). The near-monotonic trajectory of Von Neumann entropy suggests that interaction structure self-organizes more smoothly than abundance evenness, consistent with the expectation that network architecture integrates over stochastic recruitment pulses. The distinction matters because alternative stable states are defined by interaction architecture, not evenness (Petraitis 2013).

The Size 2 phase boundary

The non-monotonic pattern across clearing sizes (Fig. 1b) is perhaps the most striking result. Petraitis and Latham (1999) predicted that clearing size determines successional trajectory. The present results suggest that Size 2 clearings occupy a special position: they produced the most organized community state and showed the greatest departure from the null ($z = -22.6$). Size 2 may sit near the separatrix between basins of attraction, where the attractor landscape most strongly constrains organization. In the language of catastrophe theory, this corresponds to a cusp singularity where the system is maximally constrained by the attractor geometry (Scheffer and Carpenter 2003), consistent with Petraitis and Dudgeon's (2015a) interpretation. Alternatively, elevated entropy at larger sizes may reflect communities caught

between competing attractors. The Year × Size heatmap (Fig. S1) supports this: the deepest attractor appeared at Size 2 × 2001 (S/S_{max} = 0.683).

The elevated entropy of uncleared controls is itself noteworthy. These mature communities exhibited the highest Von Neumann entropy of any size class, possibly reflecting ongoing small-scale disturbance that sustains a more dimensionally complex interaction network. In Petraitis and Dudgeon's (2015a) catastrophe-theoretic framework, uncleared plots would occupy the cusp surface where both potential states coexist. The local increase in 8-species entropy in 2001 may reflect interannual variation in recruitment (Petraitis and Dudgeon 2015b), which can temporarily reorganize correlation structure before the successional trajectory reasserts.

Swing species and emergent modules

The bivalves' behavior as swing species is ecologically significant. In the Petraitis model, *Mytilus* beds and *Ascophyllum* canopies represent the two alternative states (Petraitis and Dudgeon 2004; Petraitis 2013). That *M. edulis*'s network allegiance shifted across strata may reflect its dual role: sometimes functioning as a dominant competitor, sometimes as a facilitator. This structural ambiguity is what one expects of a species participating in multiple stable states, echoing the role of *Mytilus* in experimental confirmation of multiple community states (Petraitis et al. 2009).

Methodological scope and limitations

The approach requires only a species-by-site abundance matrix, makes no distributional assumptions beyond Spearman correlation, and provides a single scalar capturing global

interaction organization. It connects to applications of spectral methods in quantifying network structure across disciplines (Braunstein et al. 2006), and it may be applicable wherever correlation structures carry scientific meaning.

Limitations include the temporal resolution (seven annual timepoints), which constrains dynamical inference. The 6-species subset excludes macroalgae constituting one hypothesized alternative state. Community detection used flat clustering with fixed k. Year × size interaction cells based on as few as 12 observations yield noisy correlation estimates; the heatmap (Fig. S1) should be interpreted as suggestive of spatiotemporal interaction rather than as cell-level inference. The eigendecomposition required for Von Neumann entropy scales cubically with species number; fast approximations exist for larger networks (Chen et al. 2019). Future work should apply this framework to longer-term datasets where the information-theoretic trajectory can be resolved at finer scales.

Von Neumann entropy of species correlation density matrices detected community-level structure not reliably detected by conventional diversity metrics: progressive self-organization over succession and a non-monotonic complexity landscape that pinpointed Size 2 clearings as the most organized state. These findings provide information-theoretic evidence consistent with Petraitis and colleagues' hypothesis, encoded in the relational structure between species rather than in their individual abundances.

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Declarations

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Conflicts of interest The author declares no conflict of interest.

Ethics approval This article does not contain any studies with human participants or animals performed by any of the authors.

Consent to participate Not applicable.

Consent for publication Not applicable.

Availability of data and material The intertidal succession dataset is available from Figshare (https://doi.org/10.6084/m9.figshare.3526004; Petraitis and Vidargas 2006).

Code availability Analysis code is available at https://github.com/amr28693/vn_ecology_intertidal.

Author contributions AMR conceived, designed, and executed this study and wrote the manuscript. No other person is entitled to authorship.

References

Bascompte J (2010) Structure and dynamics of ecological networks. Science 329:765–766. doi:10.1126/science.1194255

Beisner BE, Haydon DT, Cuddington K (2003) Alternative stable states in ecology. Front Ecol Environ 1:376–382. doi:10.1890/1540-9295(2003)001[0376:ASSIE]2.0.CO;2

Blanchet FG, Cazelles K, Gravel D (2020) Co-occurrence is not evidence of ecological interactions. Ecol Lett 23:1050–1063. doi:10.1111/ele.13525

Braunstein SL, Ghosh S, Severini S (2006) The Laplacian of a graph as a density matrix: a basic combinatorial approach to separability of mixed states. Ann Comb 10:291–317. doi:10.1007/s00026-006-0289-3

Brower TJ (2020) Analyzing the von Neumann entropy of contact networks. MS thesis, Utah State University, Logan

Chen PY, Wu L, Liu S, Rajapakse I (2019) Fast incremental von Neumann graph entropy computation: theory, algorithm, and applications. In: Chaudhuri K, Salakhutdinov R (eds) Proceedings of the 36th international conference on machine learning. PMLR 97:1091–1101

Connell JH (1972) Community interactions on marine rocky intertidal shores. Annu Rev Ecol Syst 3:169–192. doi:10.1146/annurev.es.03.110172.001125

Connell JH, Slatyer RO (1977) Mechanisms of succession in natural communities and their role in community stability and organization. Am Nat 111:1119–1144. doi:10.1086/283241

De Domenico M, Biamonte J (2016) Spectral entropies as information-theoretic tools for complex network comparison. Phys Rev X 6:041062. doi:10.1103/PhysRevX.6.041062

Holling CS (1973) Resilience and stability of ecological systems. Annu Rev Ecol Syst 4:1–23. doi:10.1146/annurev.es.04.110173.000245

Legendre P, Legendre L (2012) Numerical ecology, 3rd edn. Elsevier, Amsterdam

Lewontin RC (1969) The meaning of stability. Brookhaven Symp Biol 22:13–24

Magurran AE (2004) Measuring biological diversity. John Wiley & Sons

May RM (1977) Thresholds and breakpoints in ecosystems with a multiplicity of stable states. Nature 269:471–477. doi:10.1038/269471a0

Menge BA, Sutherland JP (1987) Community regulation: variation in disturbance, competition, and predation in relation to environmental stress and recruitment. Am Nat 130:730–757. doi:10.1086/284741

Minello G, Rossi L, Torsello A (2019) On the von Neumann entropy of graphs. J Complex Netw 7:491–514. doi:10.1093/comnet/cny028

Odum EP (1969) The strategy of ecosystem development. Science 164:262–270. doi:10.1126/science.164.3877.262

Paine RT (1966) Food web complexity and species diversity. Am Nat 100:65–75. doi:10.1086/282400

Passerini F, Severini S (2009) Quantifying complexity in networks: the von Neumann entropy. Int J Agent Technol Syst 1:58–67. doi:10.4018/jats.2009071005

Petraitis PS (2013) Multiple stable states in natural ecosystems. Oxford University Press, Oxford

Petraitis PS, Dudgeon SR (2004) Detection of alternative stable states in marine communities. J Exp Mar Biol Ecol 300:343–371. doi:10.1016/j.jembe.2003.12.022

Petraitis PS, Dudgeon SR (2005) Divergent succession and implications for alternative states on rocky intertidal shores. J Exp Mar Biol Ecol 326:14–26. doi:10.1016/j.jembe.2005.05.013

Petraitis PS, Dudgeon SR (2015a) Cusps and butterflies: multiple stable states in marine systems as catastrophes. Mar Freshw Res 67:37–46. doi:10.1071/MF14229

Petraitis PS, Dudgeon SR (2015b) Variation in recruitment and the establishment of alternative community states. Ecology 96:3186–3196. doi:10.1890/14-2107.1

Petraitis PS, Hoffman C (2010) Multiple stable states and relationship between thresholds in processes and states. Mar Ecol Prog Ser 413:189–200. doi:10.3354/meps08691

Petraitis PS, Latham RE (1999) The importance of scale in testing the origins of alternative community states. Ecology 80:429–442. doi:10.1890/0012-9658(1999)080[0429:TIOSIT]2.0.CO;2

Petraitis PS, Methratta ET, Rhile EC, Vidargas NA, Dudgeon SR (2009) Experimental confirmation of multiple community states in a marine ecosystem. Oecologia 161:139–148. doi:10.1007/s00442-009-1350-9

Petraitis PS, Vidargas N (2006) Marine intertidal organisms found in experimental clearings on sheltered shores, Gulf of Maine, USA. Ecology 87:796. doi:10.1890/05-0232a

Proulx SR, Promislow DEL, Phillips PC (2005) Network thinking in ecology and evolution. Trends Ecol Evol 20:345–353. doi:10.1016/j.tree.2005.04.004

Scheffer M, Carpenter S, Foley JA, Folke C, Walker B (2001) Catastrophic shifts in ecosystems. Nature 413:591–596. doi:10.1038/35098000

Scheffer M, Carpenter SR (2003) Catastrophic regime shifts in ecosystems: linking theory to observation. Trends Ecol Evol 18:648–656. doi:10.1016/j.tree.2003.09.002

Schröder A, Persson L, De Roos AM (2005) Direct experimental evidence for alternative stable states: a review. Oikos 110:3–19. doi:10.1111/j.0030-1299.2005.13962.x

Sutherland JP (1974) Multiple stable points in natural communities. Am Nat 108:859–873. doi:10.1086/282961

von Neumann J (1927) Thermodynamik quantenmechanischer Gesamtheiten. Nachr Ges Wiss Göttingen 1927:273–291

Tables

Table 1 Von Neumann entropy of the 6-species correlation density matrix over successional time (1996–2002). S_{vn}: Von Neumann entropy (bits). S/S_{max}: normalized entropy (0–1). CI: 95% bootstrap confidence interval (1,000 site-resampled replicates). Match: pairwise agreement between data-driven and a priori clustering. Cluster notation: species separated by │ are in different groups

Year	n	S_vn	S/S_max	CI lo	CI hi	Match	Clusters
1996	48	2.426	0.939	2.191	2.454	0.40	LL NL MM │ LO TT ME
1997	120	2.470	0.955	2.339	2.511	0.67	LL ME MM │ LO NL TT
1998	120	2.367	0.916	2.229	2.420	0.47	LL ME │ LO NL TT MM
1999	60	2.322	0.898	2.182	2.384	0.47	LL ME │ LO NL TT MM
2000	120	2.305	0.892	2.112	2.425	0.40	LL LO ME │ NL TT MM
2001	60	2.284	0.883	2.006	2.426	0.47	LL ME │ LO NL TT MM
2002	60	2.206	0.853	1.891	2.365	0.47	LL ME │ LO NL TT MM

Table 2 Species-label permutation tests (1,000 replicates). Species labels were shuffled within each row, destroying correlation structure while preserving marginal abundance distributions. *z*: standardized departure of observed S_{vn} from null distribution (negative = more organized than expected). Significance: *** $P < 0.001$

Stratum	**Obs.**	**Null mean**	**Null SD**	*z*	*P*
Year 1996	2.426	2.466	0.026	−1.5	0.077
Year 1997	2.470	2.533	0.010	−6.4	<0.001
Year 1998	2.367	2.546	0.010	−17.5	<0.001
Year 1999	2.322	2.523	0.022	−9.3	<0.001
Year 2000	2.305	2.532	0.010	−22.5	<0.001
Year 2001	2.284	2.511	0.021	−10.8	<0.001
Year 2002	2.206	2.520	0.021	−15.0	<0.001
Size 0	2.778	2.935	0.015	−10.2	<0.001
Size 1	2.640	2.941	0.016	−19.2	<0.001
Size 2	2.569	2.932	0.016	−22.6	<0.001
Size 4	2.705	2.941	0.015	−16.1	<0.001
Size 8	2.649	2.938	0.015	−19.1	<0.001

Table 3 Von Neumann entropy across quadrat clearing sizes. 8-species set pooled across years. Size 0 = uncleared control; *n* = 84 per size class. Column definitions as in Table 1

Size (m²)	S_vn	S/S_max	CI lo	CI hi	Match	Clusters
0	2.778	0.926	2.545	2.808	0.68	FV AN ME | LL TT | LO NL MM
1	2.640	0.880	2.387	2.727	0.61	LL ME | LO FV AN | NL TT MM
2	2.569	0.856	2.371	2.693	0.71	FV AN | LL ME | LO NL TT MM
4	2.705	0.902	2.439	2.763	0.54	AN | LL ME | LO NL TT FV MM
8	2.649	0.883	2.464	2.715	0.71	FV AN | LL ME | LO NL TT MM

Figures with legends

Fig. 1 Von Neumann entropy of species interaction structure. **a** Community self-organization over succession. Normalized Von Neumann entropy (S/S_{max}) of the 6-species (*red circles*) and 8-species (*blue squares*) correlation density matrices, 1996–2002. *Dotted line*: linear trend (6-species). *Error bars*: 95% bootstrap CIs. **b** Interaction structure along the clearing-size gradient (8-species, pooled across years). Size 0 = uncleared control. *Arrow* marks Size 2 minimum. *Error bars*: 95% bootstrap CIs

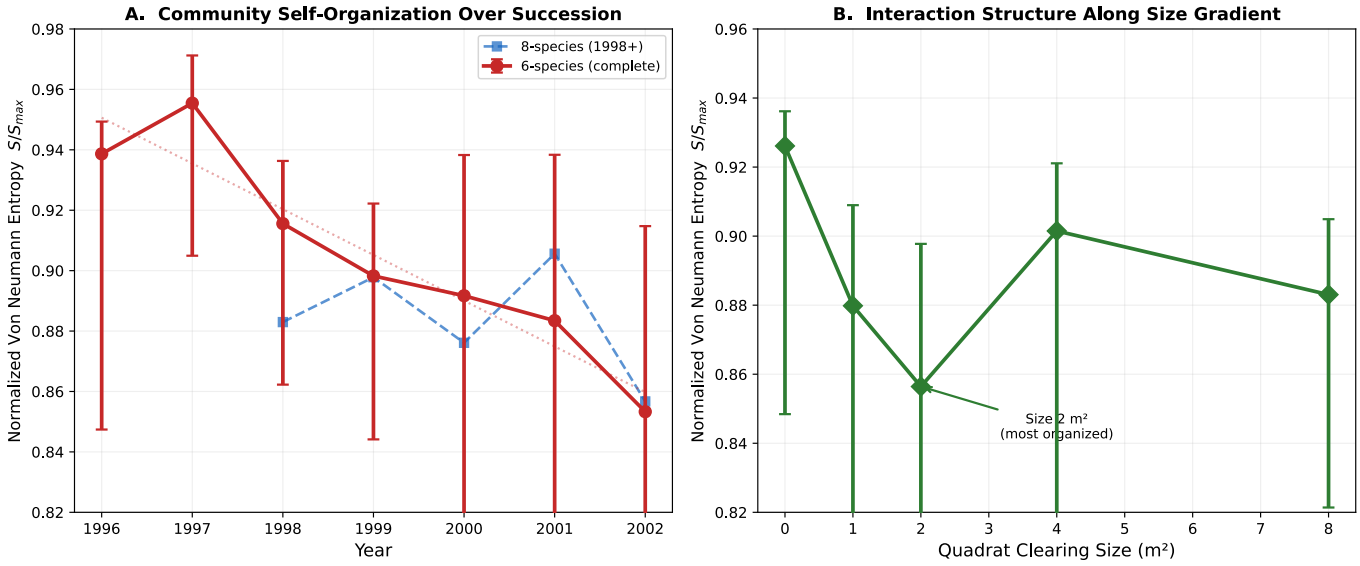
A. Community Self-Organization Over Succession
8-species (1998+)
6-species (complete)
Normalized Von Neumann Entropy S/S_{max}
Year
B. Interaction Structure Along Size Gradient
Size 2 m²
(most organized)
Normalized Von Neumann Entropy S/S_{max}
Quadrat Clearing Size (m²)

Fig. 2 Pairwise agreement between data-driven (Ward's clustering) and a priori guild groupings across strata (orange: by year; green: by size).

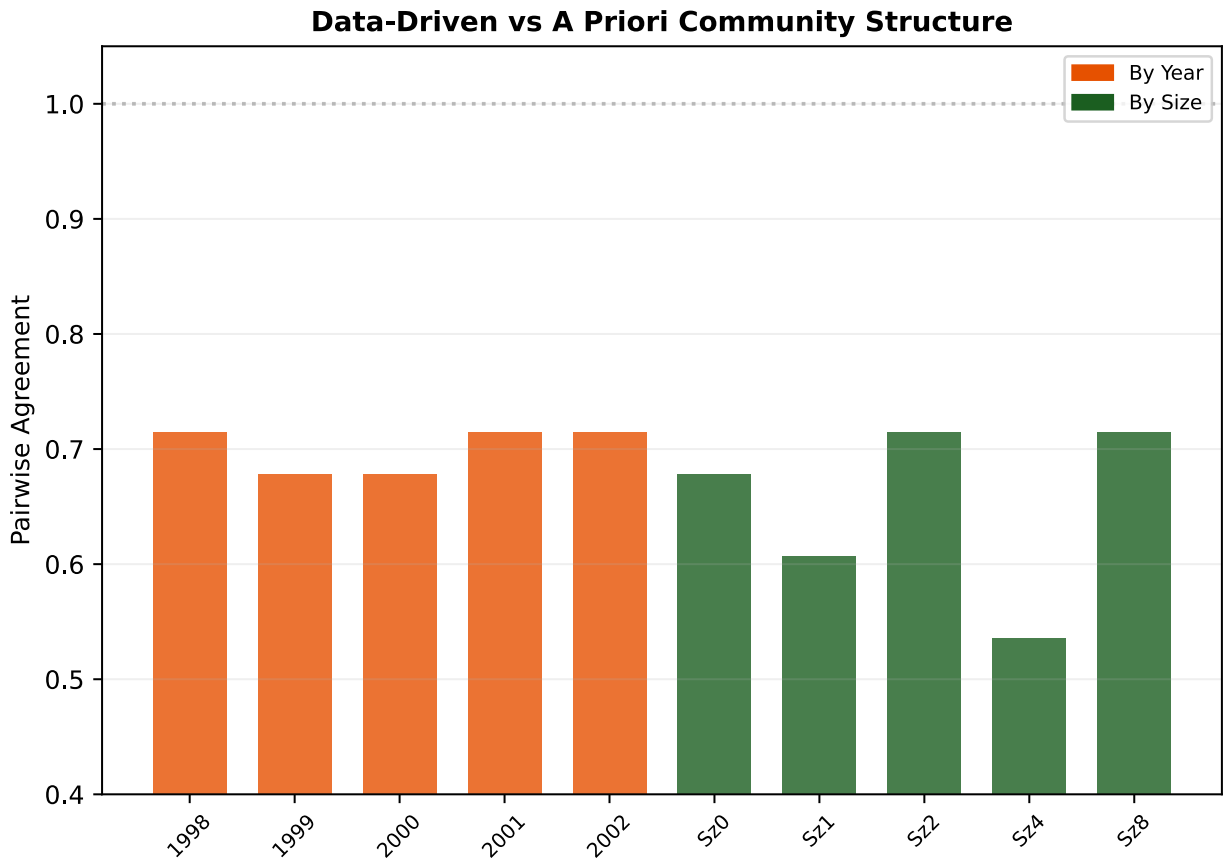
Data-Driven vs A Priori Community Structure
By Year
By Size
Pairwise Agreement
1.0
0.9
0.8
0.7
0.6
0.5
0.4
1998
1999
2000
2001
2002
Sz0
Sz1
Sz2
Sz4
Sz8

Fig. S1 Normalized Von Neumann entropy (S/Smax) across years and clearing sizes (6-species). Green: low entropy (more organized); red: high entropy (more diffuse). Dashes indicate strata with n < 8. Both gradients are visible: temporal decline (top to bottom) and the Size 2 minimum (second column). The 2001 × Size 2 cell (0.683) represents the most dimensionally reduced community state in the dataset

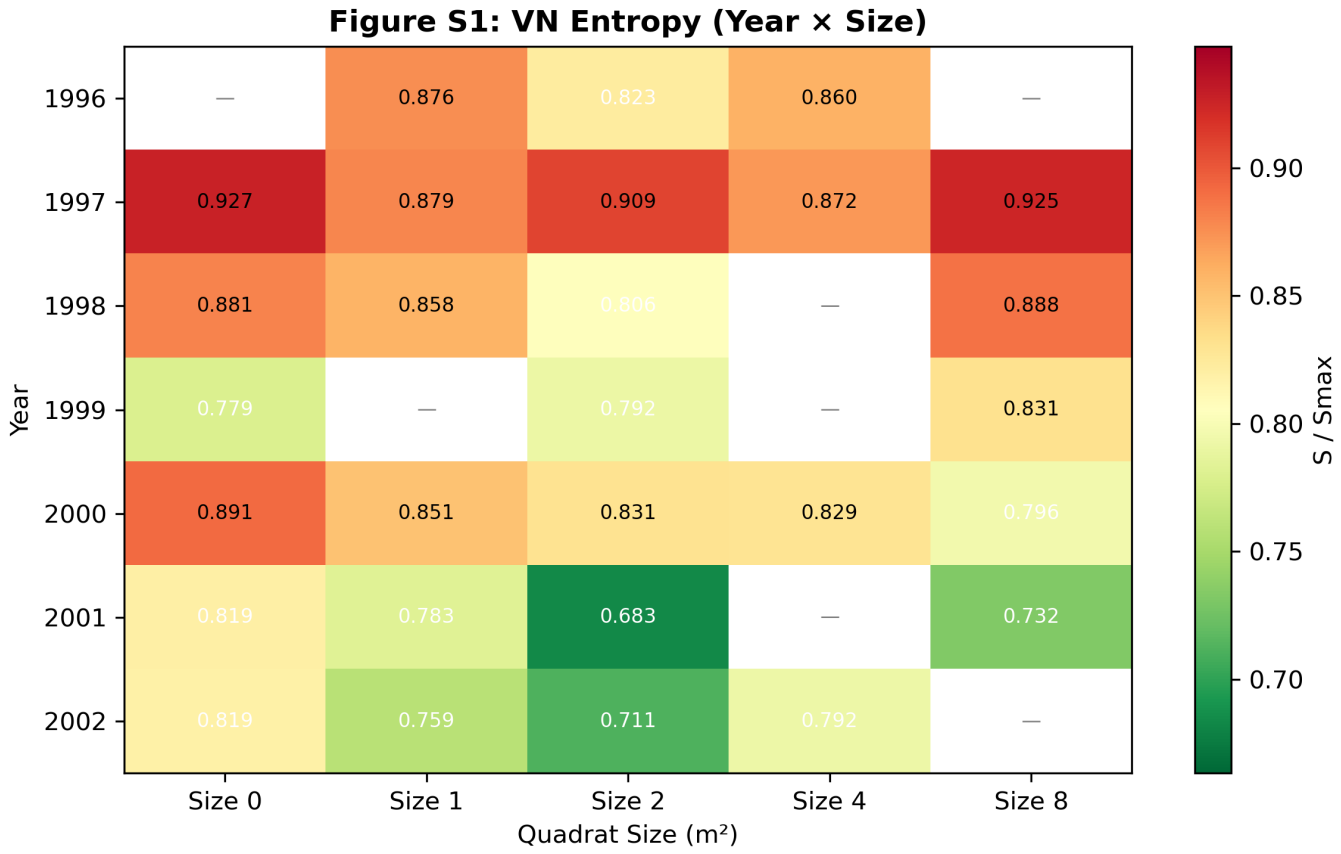
Figure S1: VN Entropy (Year × Size)
Year
1996
1997
1998
1999
2000
2001
2002
—
0.876
0.823
0.860
—
0.927
0.879
0.909
0.872
0.925
0.881
0.858
0.806
—
0.888
0.779
—
0.792
—
0.831
0.891
0.851
0.831
0.829
0.796
0.819
0.783
0.683
—
0.732
0.819
0.759
0.711
0.792
—
Size 0
Size 1
Size 2
Size 4
Size 8
Quadrat Size (m²)
0.90
0.85
0.80
0.75
0.70
S / Smax

415